

Vector Calculus

Physical Intuitions: Gradient, Divergence and Jacobian

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Tuesday

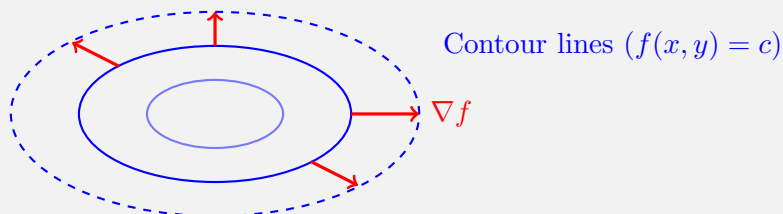
Learning Objective for Today

Understand the physical interpretation of the gradient and divergence, and master the creation of the Jacobian matrix.

Intuition 1: Gradient (∇f)

The gradient is a vector that, at any point in space, points in the direction of the **steepest ascent** of the function's value (e.g., temperature, altitude). The length of the gradient vector tells us how steep this slope is.

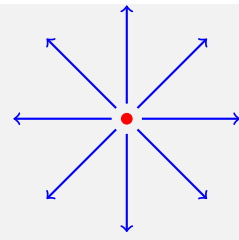
- **Data Analysis / ML:** In algorithm optimization, we are interested in descending along the gradient (Gradient Descent). If the cost function is a mountain, the gradient points to where the peak is, so we move in the *opposite* direction ($-\nabla f$) to find the minimum error.
- **Geometry:** The gradient vector is always **perpendicular** (orthogonal) to the contour lines (lines of constant value).



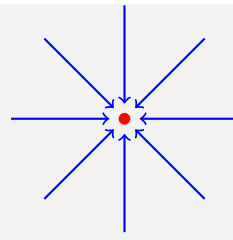
Intuition 2: Divergence ($\text{div}\mathbf{F}$ or $\nabla \cdot \mathbf{F}$)

While the gradient applies to a scalar function (e.g., a temperature field), divergence describes a **vector field** (e.g., a fluid velocity field). It tells us whether a given point behaves like a source (produces fluid) or a sink (absorbs fluid).

- **Medical Physics / Biomechanics:** Imagine modeling blood flow. If $\text{div}\mathbf{F} = 0$ (zero divergence), blood is an incompressible fluid – whatever flows into a node must flow out. If $\text{div}\mathbf{F} > 0$, we have a source. If $\text{div}\mathbf{F} < 0$, we have compression (e.g., a blockage/sink).



$\nabla \cdot \mathbf{F} > 0$ (Source)



$\nabla \cdot \mathbf{F} < 0$ (Sink)

Theory: Jacobian Matrix (Jacobian)

For a vector-valued function of multiple variables $F(x_1, \dots, x_n) = (f_1, \dots, f_m)$, the Jacobian matrix collects all the first-order partial derivatives. Intuition: The Jacobian is a **local approximation of a nonlinear function using a linear transformation**. It tells us how the spatial grid deforms (stretches, rotates) near a given point.

$$J_F = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Materials

Source: Khan Academy

Personal notes from the videos ("Gradient intuition", "Divergence intuition", "The Jacobian matrix"):

Quiz: Computational Task

Calculate the Jacobian matrix from memory for the following simple 2D nonlinear functions.

Remember: Each row of the matrix is the gradient of one component of the vector-valued function.

Task 1

Given the vector-valued function $F(x, y) = \begin{bmatrix} u(x, y) \\ v(x, y) \end{bmatrix} = \begin{bmatrix} x^2 y \\ 5x + \sin(y) \end{bmatrix}$.

Determine the Jacobian matrix J_F .

Task 2

Given the transformation from polar to Cartesian coordinates: $T(r, \theta) = \begin{bmatrix} r \cos(\theta) \\ r \sin(\theta) \end{bmatrix}$.

Determine the Jacobian matrix J_T . Then calculate the determinant of this matrix (this is the so-called scalar *Jacobian*, crucial when changing variables in double integrals).

Answer Key (to check later):

Task 1: $J_F = \begin{bmatrix} 2xy & x^2 \\ 5 & \cos(y) \end{bmatrix}$

Task 2: $J_T = \begin{bmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{bmatrix}$. Determinant: $\det(J_T) = r(\cos^2 \theta + \sin^2 \theta) = r$.